# Realized Volatility Forecasting with Time Series Foundation Models

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**Abstract**

Accurate forecasting of realized volatility is central to risk management, derivative pricing, and portfolio allocation. No study has compared multiple zero-shot time series foundation models against standard econometric benchmarks for realized volatility using formal statistical testing. We evaluate four TSFMs—Chronos-Bolt, Moirai 2.0, Lag-Llama, and Kronos—against six econometric specifications (HAR, HAR-J, HAR-RS, HARQ, Log-HAR, ARFIMA) on 50 assets from the VOLARE dataset (equities, FX, futures; 2015–2026) at horizons of 1, 5, and 22 days, using Diebold–Mariano tests and Model Confidence Sets. The best TSFMs match or exceed the best econometric benchmarks, but rankings depend on loss function, horizon, and architecture. At $h = 1$, Log-HAR leads on QLIKE while Moirai 2.0 dominates MSE and $R^2_{\text{OOS}}$; at $h \geq 5$, Moirai 2.0 leads on all metrics. TSFM performance is heterogeneous: Kronos is excluded from the MCS for virtually all assets, while Lag-Llama enters for 95% at $h = 22$ but 0% at $h = 1$. Rankings shift across asset classes, with Chronos-Bolt outperforming on FX and futures. For covariance forecasting, element-wise TSFMs achieve competitive accuracy but fail to produce viable portfolios at high dimensions, where only structurally constrained approaches succeed.

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# 1 Introduction

Volatility is a first-order input to option pricing, value-at-risk estimation, and portfolio optimization. Since the work of Andersen and Bollerslev (1998) and Barndorff-Nielsen and Shephard (2002), realized volatility constructed from high-frequency returns has served as the standard model-free measure of ex-post price variation. Forecasting realized volatility is now a central problem in financial econometrics, with direct implications for risk management, hedging, and capital allocation.

The Heterogeneous Autoregressive (HAR) model of Corsi (2009) established itself as the dominant benchmark. Its three-component structure—aggregating past realized volatility at daily, weekly, and monthly frequencies—provides a parsimonious approximation to the long-memory dynamics that characterize volatility. Subsequent extensions have incorporated jump variation (Andersen et al., 2007), semivariance asymmetries (Patton and Sheppard, 2015), and measurement error corrections (Bollerslev et al., 2016), yet the practical gains over the baseline specification have been mixed (Clements and Preve, 2021; Branco et al., 2024).

Machine learning methods—including random forests (Luong and Dokuchaev, 2018), neural networks (Bucci, 2020), and convolutional architectures (Moreno-Pino and Zohren, 2024)—have also been applied to realized volatility forecasting. Results are inconsistent: Bucci (2020) reports gains over HAR, while Branco et al. (2024) finds that linear models are difficult to outperform once the comparison accounts for multiple testing. A broader assessment by Christensen et al. (2023) concludes that machine learning offers incremental improvements but no paradigm shift for volatility prediction.

A different class of models has emerged over the past two years: time series foundation models (TSFMs), large pretrained transformers trained on billions of time series observations across domains. Leading examples include Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), and Moirai (Woo et al., 2024b), each of which has since released second-generation versions with improved architectures and expanded training data (Ansari et al.,

2025; Liu et al., 2025a). These models can produce forecasts for previously unseen time series without any retraining—a capability known as zero-shot forecasting. Several surveys document the rapid pace of development in this area (Liang et al., 2024; Ye et al., 2024; Miller et al., 2024).

Applications of TSFMs to financial time series remain limited. Goel et al. (2025) tested TimesFM on realized volatility for 20 stocks and found that fine-tuning was necessary for the model to compete with HAR; zero-shot performance was poor. Rahimikia et al. (2025) evaluated Chronos, TimesFM, Moirai, and MOMENT on daily stock returns and reported uniformly negative zero-shot results, with simple ARIMA models dominating. However, realized volatility differs from returns in ways that may favor foundation models: it is strictly positive, exhibits mean reversion, displays long-memory dependence, and follows a distribution much closer to the macroeconomic and physical series on which TSFMs were pretrained. No study has yet evaluated multiple foundation models on realized volatility with formal statistical testing.

This paper makes five contributions. First, we provide the first multi-model TSFM comparison for realized volatility forecasting, evaluating four TSFMs (Chronos-Bolt, Moirai 2.0, Lag-Llama, and Kronos) against six econometric benchmarks across three asset classes. Second, we test these models in a pure zero-shot setting—no domain-specific training—to isolate the value of general time series pretraining. Third, we analyze three forecast horizons ($h = 1, 5, 22$ days) across 50 assets, uncovering horizon-dependent and asset-class-dependent patterns in relative model performance. Fourth, we apply Model Confidence Set analysis (Hansen et al., 2011) to control for multiple comparisons, moving beyond pairwise tests. Fifth, we use the recently released VOLARE dataset (Cipollini et al., 2026), which provides methodologically consistent realized measures across U.S. equities, FX, and futures, ensuring that cross-asset comparisons are not confounded by measurement differences.

Our main findings are as follows. First, zero-shot TSFMs are competitive with the best econometric benchmarks for realized volatility forecasting—a result that stands in contrast to

prior negative findings on daily returns (Rahimikia et al., 2025) and the zero-shot evaluation in Goel et al. (2025). Second, the ranking of models depends on the loss function: Log-HAR achieves the lowest QLIKE at the one-day horizon for equities, but TSFMs dominate on MSE, MAE, and out-of-sample $R^2$ at all horizons. Moirai 2.0 achieves the lowest QLIKE at the 5-day and 22-day horizons. Third, Model Confidence Set analysis shows that Moirai 2.0 enters the 90% MCS for every equity at $h = 5$ and for 97.5% of equities at $h = 22$, while Log-HAR enters for 90% at $h = 1$. Fourth, model rankings are not stable across asset classes: Chronos-Bolt performs best for FX and futures, where volatility dynamics are less persistent, while Moirai 2.0 dominates for equities. Fifth, levels-based HAR extensions (HAR-RS, HARQ) produce negative forecasts that inflate QLIKE, a practical limitation that log-space and TSFM approaches avoid.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and sample construction. Section 4 presents the econometric and foundation model specifications, along with the forecast evaluation framework. Section 5 reports the empirical results. Section 6 contains robustness checks, and Section 7 concludes.

# 2 Literature Review

## 2.1 Realized Volatility Modeling

The theory of realized volatility originates with Andersen and Bollerslev (1998) and Andersen et al. (2001), who showed that the sum of squared intraday returns provides a consistent, nonparametric estimator of the integrated variance of asset prices. Barndorff-Nielsen and Shephard (2002) formalized this result in a semimartingale framework, establishing that realized variance converges to integrated variance as the sampling frequency increases. Subsequent work identified the key stylized facts of realized volatility: approximate log-normality, long-memory dependence with a fractional integration parameter $d \approx 0.4$, and slow mean

reversion (Andersen et al., 2003).

At ultra-high sampling frequencies, microstructure noise from bid-ask bounce and discrete price changes biases the realized variance estimator upward. Barndorff-Nielsen et al. (2008) introduced the realized kernel as a noise-consistent alternative that remains valid in the presence of market microstructure effects.

The HAR model of Corsi (2009) became the standard forecasting benchmark for realized volatility. By including daily, weekly, and monthly RV components as regressors, the model approximates long-memory behavior through a cascade of time scales corresponding to heterogeneous trading horizons. The specification is parsimonious—three regressors plus a constant—yet achieves accuracy comparable to far more complex alternatives.

A large body of work has extended the HAR framework. Andersen et al. (2007) and Barndorff-Nielsen and Shephard (2004) separated realized variance into continuous and jump components using bipower variation, yielding the HAR-J model. Patton and Sheppard (2015) and Barndorff-Nielsen et al. (2010) decomposed realized variance into positive and negative semivariance to capture asymmetric responses to upside and downside moves. Bollerslev et al. (2016) introduced the HARQ model, which interacts the daily RV regressor with realized quarticity to account for time-varying measurement error. Clements and Preve (2021) provides a practical guide to implementing and comparing these extensions.

Long-memory models offer a complementary approach. ARFIMA models (Baillie et al., 1996; Izzeldin et al., 2019) explicitly parameterize the fractional integration order, while the rough volatility framework of Gatheral et al. (2018) models volatility as a fractional Brownian motion with Hurst exponent $H \approx 0.1$. Bennedsen et al. (2022) provides cross-asset evidence on the roughness of realized volatility, though the debate over whether roughness is genuine or an artifact of microstructure noise continues.

Machine learning methods have been applied to realized volatility forecasting with mixed results. Bucci (2020) reports that feedforward and recurrent neural networks outperform HAR on a sample of U.S. stocks. Luong and Dokuchaev (2018) finds gains from random

forests. Moreno-Pino and Zohren (2024) introduces DeepVol, a convolutional neural network that processes raw order book data. In contrast, Branco et al. (2024) concludes that simple linear models are difficult to beat after correcting for multiple testing, and Christensen et al. (2023) finds that machine learning offers only incremental improvements over standard econometric methods. Recent surveys by Gunnarsson et al. (2024) and Leushuis and Petkov (2026) cover this literature.

## 2.2 Time Series Foundation Models

Time series foundation models are large pretrained models that produce forecasts for arbitrary time series without task-specific training, analogous to large language models for text (Liang et al., 2024). These models are trained on vast corpora of time series drawn from diverse domains—weather, energy, retail, transport, macroeconomics—and learn general temporal patterns that transfer to unseen series at inference time.

Three architectural families dominate the first generation of TSFMs. Chronos (Ansari et al., 2024) converts continuous values into discrete tokens via quantile binning and uses a T5 encoder-decoder architecture for autoregressive forecasting. TimesFM (Das et al., 2024) employs a patch-based decoder-only transformer with 200 million parameters, processing input in fixed-length patches and generating forecasts autoregressively. Moirai (Woo et al., 2024b) handles arbitrary numbers of variates through an "any-variate" attention mechanism and uses mixture distribution outputs for uncertainty quantification.

Second-generation models appeared in 2025 with marked improvements. Ansari et al. (2025) introduced Chronos-2 and Chronos-Bolt: the former adds multivariate support and covariate handling, while the latter is a distilled encoder-only variant that produces direct quantile forecasts and runs up to 250 times faster than the original. Moirai 2.0 (Liu et al., 2025a) demonstrated that smaller, better-trained models can match or exceed their larger predecessors—a "less is more" finding—using multi-token prediction and improved tokenization. Other models include Lag-Llama (Rasul et al., 2024), MOMENT (Goswami

et al., 2024), Timer (Liu et al., 2024), and Sundial (Liu et al., 2025b), each exploring different design choices for temporal representation and output distribution.

Standardized benchmarks have been developed to evaluate TSFMs across domains. GIFT-Eval (Aksu et al., 2024) provides a unified evaluation protocol across multiple datasets and forecast horizons. FEV-Bench (Shchur et al., 2025) focuses on forecasting evaluation with emphasis on calibration. TSFM-Bench (Li et al., 2024) compares models across zero-shot and fine-tuned settings. A consistent finding is that no single model dominates across all domains and horizons, which motivates domain-specific evaluations such as the one we undertake here. Adler et al. (2025) raises additional concerns about the calibration of TSFM prediction intervals, finding systematic miscoverage across several models.

## 2.3 Foundation Models for Volatility Forecasting

The application of TSFMs to volatility forecasting is recent and limited. The closest prior work is Goel et al. (2025), who tested TimesFM 1.0 on realized volatility for 20 stocks. They found that zero-shot TimesFM underperformed HAR and that fine-tuning was necessary to achieve competitive accuracy. Our paper differs in several respects: we evaluate two foundation models from different families (Chronos-Bolt and Moirai 2.0), use newer model generations, and apply Model Confidence Set analysis to control for multiple comparisons. The same research group also applied TimesFM to Value-at-Risk forecasting (Goel et al., 2024).

Rahimikia et al. (2025) evaluated Chronos, TimesFM, Moirai, and MOMENT on daily stock returns and found that zero-shot TSFMs consistently underperformed ARIMA. However, daily returns differ from realized volatility in fundamental ways: returns are approximately symmetric, near-white-noise, and lack the strong serial dependence that characterizes realized volatility. Realized volatility is strictly positive, mean-reverting, and displays long-memory dynamics—properties that bring it closer to the macroeconomic and physical time series on which TSFMs were pretrained. This distinction suggests that negative results on

returns need not carry over to realized measures.

Other financial applications include stock price forecasting (Łaniewski and Ślepaczuk, 2025; Valeyre and Aboura, 2024), foreign exchange volatility modeling (Nguyen et al., 2025), and multivariate financial time series (Marconi, 2025). Finance-specific foundation models have also been proposed, including FinCast (Zhu et al., 2025) and the finance-oriented Kronos (Shi et al., 2025), though neither is publicly available at the time of writing. We include the general-purpose Kronos-T5 model (Woo et al., 2024a) (distinct from the finance-specific variant) in our evaluation alongside three other publicly available TSFMs.

The question of whether to fine-tune or apply TSFMs zero-shot is itself an active area of research. Das et al. (2025) explored in-context fine-tuning for TimesFM, while Germán-Morales et al. (2024) and Gupta et al. (2024) investigated parameter-efficient adaptation via LoRA. Testing zero-shot performance first establishes an important baseline: if a TSFM already performs well without adaptation, the marginal value of fine-tuning may be limited, and the zero-shot workflow offers practical advantages in simplicity and computational cost.

We contribute to this emerging literature by providing the first evaluation that compares multiple time series foundation models against a broad set of econometric benchmarks for realized volatility, with formal statistical testing via Diebold–Mariano tests and Model Confidence Sets.

# 3 Data and Sample Construction

## 3.1 The VOLARE Dataset

We use the VOLARE (VOLatility Archive for Realized Estimates) dataset of Cipollini et al. (2026), which provides daily realized volatility measures for a broad cross-section of financial assets. VOLARE is constructed from ultra-high-frequency tick data sourced from Kibot, covering 40 U.S. equities, 5 major currency pairs, and 5 commodity and index futures contracts. The equity sample begins on January 2, 2015 and runs through January 30, 2026,

yielding 2,786 trading days per stock. The FX and futures samples begin on September 25, 2009 (4,242 and 4,228 trading days, respectively).

For each asset-day, VOLARE supplies a suite of realized measures computed at multiple sampling frequencies. We use the 5-minute realized variance ($RV^{(5)}$) as our primary target, along with 5-minute bipower variation ($BPV^{(5)}$), positive and negative realized semivariance ($RS^{+(5)}$, $RS^{-(5)}$), and 5-minute realized quarticity ($RQ^{(5)}$). All measures are expressed in decimal squared returns—for instance, a typical daily $RV^{(5)}$ for a U.S. equity is approximately $2 \times 10^{-4}$, corresponding to an annualized volatility of roughly 22%.

Three features distinguish VOLARE from earlier datasets such as the Oxford-Man realized library. First, the realized measures are constructed using a uniform methodology across all assets, eliminating cross-asset inconsistencies that arise when measures are drawn from heterogeneous sources. Second, the dataset includes subsampled estimators ($RV_{\text{SS}}^{(5)}$), realized kernels, and range-based volatility estimators alongside the standard measures, enabling robustness checks across estimator types. Third, the multi-asset-class coverage—equities, currencies, and futures—permits tests of whether model rankings are stable across markets with different microstructure characteristics and volatility dynamics.

## 3.2 Sample Selection and Descriptive Statistics

Our equity sample comprises the 40 stocks in VOLARE that span the full 2015–2026 period: AAPL, ADBE, AMD, AMGN, AMZN, AXP, BA, CAT, CRM, CSCO, CVX, DIS, GE, GOOGL, GS, HD, HON, IBM, JNJ, JPM, KO, MCD, META, MMM, MRK, MSFT, NFLX, NKE, NVDA, ORCL, PG, PM, SHW, TRV, TSLA, UNH, V, VZ, WMT, and XOM. These cover all 11 GICS sectors and include both mega-cap (AAPL, MSFT) and mid-cap (SHW, PM) names. The FX sample consists of five major currency pairs (AUDUSD, EURUSD, GBPUSD, USDCAD, USDJPY) and the futures sample covers five contracts (Corn, Crude Oil, E-mini S&P 500, Gold, Natural Gas).

Table 1 reports descriptive statistics for the 5-minute realized variance across the three

asset classes. Several patterns stand out. First, realized variance is strongly right-skewed across all asset classes, with kurtosis values ranging from 75 (CRM) to over 1,000 (JNJ), confirming that extreme volatility episodes are a persistent feature of the data. Second, the first-order autocorrelation of daily RV averages 0.606 for equities, consistent with the well-documented persistence of volatility. The 22-day autocorrelation averages 0.085, indicating slow decay. Third, the FX pairs display RV approximately two orders of magnitude smaller than equities ($\bar{RV} \approx 3 \times 10^{-5}$ vs. $2 \times 10^{-4}$), with comparable persistence ($\bar{\rho}_1 = 0.52$). Fourth, futures exhibit the widest cross-asset heterogeneity: the E-mini S&P 500 (ES) has $\rho_1 = 0.779$, while Gold (GC) shows near-zero autocorrelation ($\rho_1 = -0.001$), presenting a natural stress test for forecasting models.

Table 1: Descriptive statistics for daily 5-minute realized variance ($RV^{(5)}$). The equity panel reports cross-sectional averages across 40 stocks; FX and futures report individual assets. $\rho_k$ denotes the sample autocorrelation at lag $k$. All RV values are in decimal squared returns ($\times 10^{-4}$).

| | $N$ | Mean | Median | Skew | Kurt | $\rho_1$ | $\rho_{22}$ |
|---|---|---|---|---|---|---|---|
| *Panel A: Equities (cross-sectional average, 40 stocks)* | | | | | | | |
| Average | 2,786 | 1.91 | 1.12 | 13.5 | 269 | 0.606 | 0.085 |
| *Panel B: Foreign Exchange (5 pairs)* | | | | | | | |
| AUDUSD | 4,241 | 0.48 | 0.37 | — | — | 0.577 | — |
| EURUSD | 4,241 | 0.28 | 0.22 | — | — | 0.570 | — |
| GBPUSD | 4,242 | 0.32 | 0.24 | — | — | 0.484 | — |
| USDCAD | 4,240 | 0.25 | 0.19 | — | — | 0.654 | — |
| USDJPY | 4,239 | 0.33 | 0.22 | — | — | 0.325 | — |
| *Panel C: Futures (5 contracts)* | | | | | | | |
| Corn (C) | 4,185 | 2.48 | 1.54 | — | — | 0.178 | — |
| Crude Oil (CL) | 4,223 | 6.87 | 3.87 | — | — | 0.195 | — |
| E-mini S&P (ES) | 4,224 | 1.11 | 0.53 | — | — | 0.779 | — |
| Gold (GC) | 4,206 | 3.63 | 2.32 | — | — | −0.001 | — |
| Natural Gas (NG) | 4,221 | 10.30 | 6.35 | — | — | 0.187 | — |

# 4 Methodology

## 4.1 Econometric Benchmarks

We consider six econometric specifications that span the main approaches to realized volatility forecasting: the HAR family and its extensions, which exploit multi-scale aggregation, and ARFIMA, which explicitly models long memory.

**HAR Model.** The Heterogeneous Autoregressive model of Corsi (2009) captures the multi-scale persistence of realized volatility by aggregating past RV at daily, weekly, and monthly frequencies:

$$RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \varepsilon_{t+h}, \tag{1}$$

where $RV_{t-k:t} = k^{-1} \sum_{i=0}^{k-1} RV_{t-i}$ denotes the average realized volatility over the previous $k$ days. This three-component structure approximates long-memory behavior through the actions of heterogeneous market participants—short-term traders, medium-term portfolio managers, and long-term institutional investors—who operate at different time horizons (Corsi, 2009).

**HAR-J Model.** The HAR-J model augments the baseline HAR with a jump component to separate continuous and discontinuous variation (Andersen et al., 2007):

$$RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \beta_J J_t + \varepsilon_{t+h}, \tag{2}$$

where $J_t = \max(RV_t - BPV_t, 0)$ measures the jump component as the positive part of the difference between realized variance and bipower variation (Barndorff-Nielsen and Shephard, 2004). A negative coefficient on $J_t$ indicates that large jumps reduce, rather than increase, future volatility.

**HAR-RS Model.** The HAR-RS model decomposes realized variance into positive and negative semivariance components to capture asymmetric volatility responses (Patton and Sheppard, 2015):

$$RV_{t+h} = \beta_0 + \beta_1^+ RS_t^+ + \beta_1^- RS_t^- + \beta_2^+ RS_{t-5:t}^+ + \beta_2^- RS_{t-5:t}^- + \beta_3^+ RS_{t-22:t}^+ + \beta_3^- RS_{t-22:t}^- + \varepsilon_{t+h}, \quad (3)$$

where $RS_t^+$ and $RS_t^-$ denote good and bad realized semivariance, respectively, and $RS_t^+ + RS_t^- = RV_t$ by construction (Barndorff-Nielsen et al., 2010). The six-regressor structure allows upside and downside risk to follow separate dynamics at each aggregation frequency.

**HARQ Model.** The HARQ model of Bollerslev et al. (2016) accounts for time-varying measurement error in realized volatility by interacting the daily RV regressor with the square root of realized quarticity:

$$RV_{t+h} = \beta_0 + (\beta_1 + \beta_1^Q \sqrt{RQ_t})\, RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \varepsilon_{t+h}, \quad (4)$$

where $RQ_t$ is the realized quarticity. The interaction term attenuates the daily RV signal when measurement noise is high, as indicated by large values of $RQ_t$.

**Log-HAR Model.** The Log-HAR model applies the HAR specification to log-transformed realized volatility (Andersen et al., 2003):

$$\log RV_{t+h} = \beta_0 + \beta_1 \log RV_t + \beta_2 \log RV_{t-5:t} + \beta_3 \log RV_{t-22:t} + \varepsilon_{t+h}. \quad (5)$$

The logarithmic transformation brings the RV distribution closer to normality, improving OLS efficiency. Point forecasts in levels are recovered via bias-corrected retransformation: $\widehat{RV}_{t+h} = \exp(\hat{y}_{t+h} + \hat{\sigma}^2/2)$, where $\hat{\sigma}^2$ is the estimated residual variance.

**ARFIMA Model.** The ARFIMA model captures the well-documented long-memory property of realized volatility through fractional differencing (Baillie et al., 1996):

$$(1-L)^d\,\Phi(L)\,(\log RV_t - \mu) = \Theta(L)\,\varepsilon_t, \tag{6}$$

where $d \in (0, 0.5)$ is the fractional integration parameter, $\Phi(L)$ and $\Theta(L)$ are autoregressive and moving average lag polynomials of orders $p$ and $q$ (with $p, q \leq 2$ selected by AIC), and $L$ is the lag operator. The parameter $d$ is estimated by maximum likelihood. The fractional differencing filter is truncated at $K = 100$ lags.

All HAR-family models are estimated by OLS with Newey–West heteroskedasticity and autocorrelation consistent standard errors. The ARFIMA model is re-estimated every 22 trading days to balance computational cost against parameter staleness.

We do not include the Realized GARCH model of Hansen et al. (2012) as a benchmark. Realized GARCH requires daily returns as an additional input, which VOLARE does not provide, and no publicly available Python implementation exists—the standard `arch` package supports only GARCH and EGARCH variants. Implementation via `rpy2` and the R `rugarch` package is feasible but introduces cross-language dependencies that complicate reproducibility. We note that the HAR-J and HARQ specifications already incorporate jump variation and measurement error effects that Realized GARCH targets through a different mechanism.

## 4.2 Time Series Foundation Models

We evaluate four time series foundation models (TSFMs) in a zero-shot setting—applied directly to realized volatility series without any domain-specific training or fine-tuning. Fine-tuning remains a direction for future work.[1]

[1] We originally planned to include TimesFM 2.5 (Das et al., 2024), but its publicly released model weights are incompatible with the current software package, preventing reliable inference.

**General pipeline.** Because TSFMs originate in the machine learning literature, we briefly describe the forecasting pipeline in terms familiar to econometricians. Every TSFM follows three steps: *tokenization*, *encoding*, and *decoding*. First, the raw time series—here, a vector of 512 daily RV observations—is converted into a sequence of discrete or continuous *tokens*, the model's internal representation of the data. Tokenization serves the same role as variable construction in an econometric model: it determines which features of the history the model can "see." Different TSFMs tokenize differently (quantile binning, lag features, kernel embeddings), and we describe each below. Second, the token sequence is processed by a *transformer*, a neural network architecture in which each position in the sequence attends to all other positions via learned attention weights. The attention mechanism allows the model to capture dependencies at arbitrary lags without imposing the specific functional form that HAR prescribes (daily, weekly, monthly components). The transformer's parameters—attention weights, layer norms, feed-forward coefficients—are fixed during our evaluation; they were learned during pretraining on large external corpora (tens of billions of time series observations drawn from weather, energy, retail, transport, and macroeconomic domains). Third, the transformer's output is passed through a *decoding head* that maps the internal representation back to forecasts in the original scale. Decoding heads differ across models: some output quantiles of the predictive distribution directly, others parameterize an explicit probability distribution (e.g., Student-$t$) from which forecast trajectories are sampled. In both cases, we extract the median as the point forecast, analogous to reporting the conditional median from a quantile regression.

**Chronos-Bolt.** Chronos-Bolt is a distilled variant of the Chronos family (Ansari et al., 2024, 2025). The original Chronos model tokenizes continuous values by mapping them into 4,096 discrete bins via quantile binning—each observation is replaced by the index of its nearest quantile in a reference distribution, much as one would discretize a continuous variable into histogram bins. The model then processes these token indices through a T5

encoder-decoder transformer and generates forecasts autoregressively, predicting one token at a time. Chronos-Bolt replaces this two-step architecture with an encoder-only transformer that reads the entire input context and produces all forecast quantiles in a single forward pass, without iterative decoding. This yields a 250× speedup over the original while maintaining competitive accuracy (Ansari et al., 2025). The decoding head outputs quantile forecasts directly (e.g., the 10th, 50th, and 90th percentiles), bypassing the need to sample from a token distribution. We use both the small (48M parameters) and base (205M parameters) checkpoints (`amazon/chronos-bolt-small`, `amazon/chronos-bolt-base`) and take the median (0.5 quantile) as the point forecast.

**Moirai 2.0.** Moirai 2.0 (Liu et al., 2025a) builds on the original Moirai architecture (Woo et al., 2024b). It tokenizes the input series by dividing it into fixed-length *patches*—contiguous blocks of consecutive observations (e.g., 32 days each)—and projects each patch into the transformer's embedding space via a learned linear map. This is analogous to constructing non-overlapping moving-window summaries of the series, except that the summary is learned rather than hand-specified. The transformer is decoder-only: it processes patches causally (each patch attends only to itself and earlier patches), preserving the temporal ordering. Moirai 2.0 uses *multi-token prediction*, generating multiple future patches in a single pass rather than one step at a time. The decoding head outputs nine quantile levels (0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.9, 0.95) per forecast step. We use the `Salesforce/moirai-2.0-R-small` checkpoint and extract the median quantile as the point forecast.

**Lag-Llama.** Lag-Llama (Rasul et al., 2024) is a decoder-only foundation model based on the LLaMA architecture, adapted for univariate probabilistic time series forecasting. Unlike the patch-based models above, Lag-Llama tokenizes each time step individually using a set of *lag features*: at each position $t$, the input vector consists of the current value and a fixed set of lagged values $(y_{t-1}, y_{t-2}, y_{t-7}, y_{t-14}, \ldots)$, concatenated with time-related covariates. This lag-based design is conceptually closer to an autoregressive econometric specification—the model

explicitly receives lagged values as inputs—though the transformer learns the mapping from lags to forecasts nonparametrically rather than through a linear projection. The decoding head parameterizes a Student-$t$ mixture distribution at each forecast step: it outputs the location, scale, and degrees-of-freedom parameters for each mixture component. Forecasts are generated by sampling trajectories from this distribution and taking the median across samples. We use the pretrained checkpoint (8 layers, 4 attention heads, 36-dimensional embeddings per head), sample 20 trajectories, and report the median as the point forecast.

**Kronos.** Kronos (Woo et al., 2024a) uses *kernel-based tokenization*: each continuous value is represented as a weighted combination of a set of learned kernel basis functions, producing a soft embedding rather than a hard bin assignment. The embedding is then processed by a T5 encoder-decoder transformer backbone. At decoding time, the model samples token-level outputs from the learned distribution and maps them back to continuous values through the inverse kernel transform. We use the `Salesforce/kronos-t5-tiny` checkpoint (approximately 8M parameters) with a sample count of 5 and take the median prediction as the point forecast.

**Zero-shot protocol.** All four TSFMs receive a rolling context window of 512 daily RV observations as input, with no additional features or covariates. For multi-step horizons ($h > 1$), the point forecast is the mean of the $h$-step output vector. No parameters are updated during the out-of-sample evaluation period; all forecasts are purely zero-shot. The 512-day context length is the default for these architectures, just as the 252-day rolling window is standard for HAR estimation. This reflects each approach's recommended configuration rather than an equal-information comparison; we view this as the practically relevant setup, since a practitioner would deploy each model with its standard settings. We extract the median rather than the mean of each model's predictive distribution because the median is the optimal point forecast under MAE loss and is more robust to the right-skewed forecast distributions that arise in volatility applications; for symmetric distributions, the two

coincide.

## 4.3 Forecast Evaluation

**Walk-forward design.** We employ a walk-forward evaluation scheme with a rolling origin and a fixed training window of 252 trading days (approximately one year). At each origin, the econometric model is estimated on the most recent 252 observations; the estimation window then slides forward one day at a time, producing daily re-estimated forecasts over a test block of 126 days (approximately six months). The origin then advances by 126 days and the process repeats until the end of the sample; the training window does not expand. For the equity sample (2,786 days), this produces approximately 2,408 out-of-sample forecasts per model–asset combination at $h = 1$; the FX and futures samples (4,240+ days) yield approximately 3,862. Foundation models follow the same walk-forward timing but require no estimation step; only the context window slides forward.

**Forecast horizons.** We evaluate three forecast horizons: $h = 1$ (one day), $h = 5$ (one week), and $h = 22$ (one month). The forecast target at horizon $h$ is the average realized volatility over the next $h$ days, $h^{-1} \sum_{i=1}^{h} RV_{t+i}$. All models produce direct $h$-step forecasts.

**Loss functions.** Our primary loss function is the quasi-likelihood (QLIKE) loss of Patton (2011):

$$L_{\text{QLIKE}}(\hat{\sigma}_t^2, RV_t) = \frac{RV_t}{\hat{\sigma}_t^2} - \log \frac{RV_t}{\hat{\sigma}_t^2} - 1, \tag{7}$$

where $\hat{\sigma}_t^2$ is the forecast and $RV_t$ is the realized value. Patton (2011) shows that QLIKE preserves the ranking of forecasts regardless of the choice of volatility proxy, making it the preferred metric when the true conditional variance is unobserved. We also report mean squared error (MSE), mean absolute error (MAE), and out-of-sample $R^2$ ($R^2_{\text{OOS}}$). All forecasts are floored at $10^{-6}$ to prevent numerical instability in the QLIKE calculation when models produce near-zero predictions.

**Statistical tests.** Pairwise forecast comparisons use the Diebold and Mariano (1995) test with Newey–West variance estimation, applied to QLIKE loss differentials. To address the multiple comparison problem inherent in evaluating eleven models, we compute the Model Confidence Set (MCS) of Hansen et al. (2011). The MCS identifies the subset of models whose forecasting ability cannot be statistically distinguished from the best model at a given significance level. We use the $T_{\max}$ test statistic with a block bootstrap (block length = 22 days, $B = 10{,}000$ replications) and significance level $\alpha = 0.10$.

# 5 Empirical Results

Table 2 reports average loss function values across 40 equities for each model and horizon. Table 3 presents Model Confidence Set inclusion rates. Tables 4 and 5 extend the analysis to FX and futures.

Table 2: Forecast accuracy for 40 U.S. equities (VOLARE). Values are cross-sectional averages of per-asset loss functions. Bold indicates the best value in each column within each panel. † denotes inflated QLIKE due to near-zero forecasts or systematic forecast bias.

| Model | MSE ($\times 10^{-6}$) | MAE ($\times 10^{-4}$) | QLIKE | $R^2_{\text{OOS}}$ |
|---|---|---|---|---|
| *Panel: $h = 1$ (1 day)* | | | | |
| HAR | 0.244 | 1.43 | 1.38† | 0.335 |
| HAR-J | 0.242 | 1.42 | 1.16† | 0.349 |
| HAR-RS | 0.265 | 1.52 | 3.73† | 0.268 |
| HARQ | 0.297 | 1.49 | 16.46† | 0.111 |
| Log-HAR | 0.257 | 1.31 | **0.220** | 0.238 |
| ARFIMA | 0.295 | 1.63 | 0.314 | 0.153 |
| Chronos-Bolt-S | 0.216 | 1.18 | 0.229 | 0.408 |
| Chronos-Bolt-B | 0.213 | **1.17** | 0.230 | 0.420 |
| Moirai-2.0-S | **0.210** | 1.19 | 0.226 | **0.426** |
| Lag-Llama | 0.291 | 1.65 | 0.402 | 0.127 |
| Kronos | 0.231 | 1.31 | 1.09† | 0.348 |
| *Panel: $h = 5$ (5 days)* | | | | |
| HAR | 0.163 | 1.40 | 2.70† | 0.209 |
| HAR-J | 0.159 | 1.39 | 2.47† | 0.231 |
| HAR-RS | 0.184 | 1.50 | 7.04† | 0.101 |
| HARQ | 0.192 | 1.43 | 16.16† | 0.024 |
| Log-HAR | 0.164 | 1.28 | 0.231 | 0.135 |
| ARFIMA | 0.170 | 1.44 | 0.244 | 0.178 |
| Chronos-Bolt-S | 0.127 | 1.09 | 0.213 | 0.355 |
| Chronos-Bolt-B | 0.128 | 1.09 | 0.216 | 0.346 |
| Moirai-2.0-S | **0.122** | **1.07** | **0.188** | **0.384** |
| Lag-Llama | 0.181 | 1.43 | 0.305 | 0.044 |
| Kronos | 0.143 | 1.22 | 0.867 | 0.243 |
| *Panel: $h = 22$ (22 days)* | | | | |
| HAR | 0.117 | 1.47 | 6.06† | -0.031 |
| HAR-J | 0.117 | 1.46 | 5.92† | -0.015 |
| HAR-RS | 0.130 | 1.57 | 9.40† | -0.145 |
| HARQ | 0.131 | 1.48 | 12.42† | -0.184 |
| Log-HAR | 0.115 | 1.35 | 0.350 | -0.044 |
| ARFIMA | 0.119 | 1.35 | 0.275 | -0.014 |
| Chronos-Bolt-S | 0.100 | 1.13 | 0.322 | 0.085 |
| Chronos-Bolt-B | 0.101 | 1.13 | 0.325 | 0.077 |
| Moirai-2.0-S | **0.098** | **1.07** | **0.257** | **0.110** |
| Lag-Llama | 0.109 | 1.25 | 0.283 | 0.009 |
| Kronos | 0.108 | 1.25 | 0.913 | 0.012 |

† Marks QLIKE $> 1$. For HAR-RS and HARQ, this reflects near-zero forecasts from levels-based OLS; for Kronos, it reflects systematic forecast bias.

Table 3: Model Confidence Set inclusion rates for 40 U.S. equities (VOLARE). Each cell reports the percentage of stocks for which the model is included in the MCS at the 10% significance level (QLIKE loss, $T_{\max}$ statistic, block bootstrap with $B = 10{,}000$).

| Model | $h = 1$ (%) | $h = 5$ (%) | $h = 22$ (%) |
|---|---|---|---|
| HAR | 42.5 | 42.5 | 20.0 |
| HAR-J | 17.5 | 35.0 | 20.0 |
| HAR-RS | 2.5 | 2.5 | 12.5 |
| HARQ | 0.0 | 0.0 | 7.5 |
| Log-HAR | 90.0 | 65.0 | 75.0 |
| ARFIMA | 2.5 | 30.0 | 87.5 |
| Chronos-Bolt-S | 42.5 | 30.0 | 37.5 |
| Chronos-Bolt-B | 37.5 | 25.0 | 30.0 |
| Moirai-2.0-S | 77.5 | 100.0 | 97.5 |
| Lag-Llama | 0.0 | 5.0 | 95.0 |
| Kronos | 2.5 | 0.0 | 7.5 |

Table 4: Forecast accuracy for 5 FX pairs (VOLARE). Values are averages across the five currency pairs. Format as Table 2.

| Model | MSE ($\times 10^{-8}$) | MAE ($\times 10^{-4}$) | QLIKE | $R^2_{\text{OOS}}$ |
|---|---|---|---|---|
| *Panel: $h = 1$ (1 day)* | | | | |
| HAR | 0.174 | 0.14 | 0.177 | 0.241 |
| HAR-J | 0.170 | 0.14 | 0.196 | 0.258 |
| HAR-RS | 0.192 | 0.15 | 0.249 | 0.180 |
| HARQ | 0.196 | 0.14 | 0.628 | 0.167 |
| Log-HAR | 0.174 | 0.13 | **0.163** | 0.247 |
| ARFIMA | 0.207 | 0.16 | 0.242 | 0.107 |
| Chronos-Bolt-S | 0.168 | 0.12 | 0.174 | 0.277 |
| Chronos-Bolt-B | **0.167** | **0.12** | 0.176 | **0.279** |
| Moirai-2.0-S | 0.222 | 0.19 | 0.329 | -0.008 |
| Lag-Llama | 0.207 | 0.17 | 0.305 | 0.080 |
| Kronos | 0.170 | 0.14 | 0.299 | 0.245 |
| *Panel: $h = 5$ (5 days)* | | | | |
| HAR | 0.082 | 0.12 | 0.143 | 0.284 |
| HAR-J | 0.079 | 0.12 | 0.149 | 0.305 |
| HAR-RS | 0.098 | 0.13 | 0.240 | 0.173 |
| HARQ | 0.085 | 0.12 | 0.399 | 0.259 |
| Log-HAR | 0.080 | 0.11 | **0.120** | 0.306 |
| ARFIMA | 0.098 | 0.13 | 0.178 | 0.147 |
| Chronos-Bolt-S | **0.075** | **0.10** | 0.128 | **0.353** |
| Chronos-Bolt-B | 0.075 | 0.10 | 0.129 | 0.353 |
| Moirai-2.0-S | 0.110 | 0.17 | 0.250 | -0.025 |
| Lag-Llama | 0.098 | 0.14 | 0.194 | 0.117 |
| Kronos | 0.076 | 0.12 | 0.198 | 0.311 |
| *Panel: $h = 22$ (22 days)* | | | | |
| HAR | 0.052 | 0.13 | 0.186 | 0.171 |
| HAR-J | 0.052 | 0.12 | 0.197 | 0.177 |
| HAR-RS | 0.065 | 0.13 | 0.223 | 0.004 |
| HARQ | 0.053 | 0.12 | 0.367 | 0.155 |
| Log-HAR | 0.049 | 0.12 | **0.147** | 0.205 |
| ARFIMA | 0.060 | 0.12 | 0.185 | 0.033 |
| Chronos-Bolt-S | **0.047** | **0.10** | 0.161 | **0.250** |
| Chronos-Bolt-B | 0.047 | 0.10 | 0.163 | 0.248 |
| Moirai-2.0-S | 0.064 | 0.16 | 0.212 | -0.093 |
| Lag-Llama | 0.053 | 0.13 | 0.164 | 0.119 |
| Kronos | 0.050 | 0.12 | 0.358 | 0.152 |

Table 5: Forecast accuracy for 5 futures contracts (VOLARE). Values are averages across the five contracts. Format as Table 2.

| Model | MSE ($\times 10^{-6}$) | MAE ($\times 10^{-4}$) | QLIKE | $R^2_{\text{OOS}}$ |
|---|---|---|---|---|
| *Panel: $h = 1$ (1 day)* | | | | |
| HAR | 14.818 | 2.72 | 0.225 | 0.244 |
| HAR-J | 14.958 | 2.65 | 0.707 | **0.258** |
| HAR-RS | 15.503 | 2.81 | $5.82^{\dagger}$ | 0.180 |
| HARQ | 15.156 | 2.65 | $22.34^{\dagger}$ | 0.089 |
| Log-HAR | **14.143** | 2.09 | **0.193** | 0.225 |
| ARFIMA | 16.437 | 2.95 | 0.346 | 0.009 |
| Chronos-Bolt-S | 14.335 | 1.99 | 0.213 | 0.237 |
| Chronos-Bolt-B | 14.251 | **1.95** | 0.221 | 0.246 |
| Moirai-2.0-S | 14.509 | 1.99 | 0.289 | 0.213 |
| Lag-Llama | 14.860 | 2.77 | $1.12^{\dagger}$ | 0.074 |
| Kronos | 14.387 | 2.31 | $10.57^{\dagger}$ | -2.099 |
| *Panel: $h = 5$ (5 days)* | | | | |
| HAR | 4.574 | 2.56 | 0.434 | 0.253 |
| HAR-J | 4.918 | 2.48 | 0.839 | 0.245 |
| HAR-RS | 4.615 | 2.60 | $4.07^{\dagger}$ | 0.161 |
| HARQ | 4.792 | 2.49 | $22.10^{\dagger}$ | 0.181 |
| Log-HAR | 4.442 | 1.98 | **0.165** | 0.226 |
| ARFIMA | 6.262 | 2.70 | 0.269 | -0.040 |
| Chronos-Bolt-S | **4.352** | 1.80 | 0.185 | 0.273 |
| Chronos-Bolt-B | 4.384 | **1.77** | 0.196 | **0.275** |
| Moirai-2.0-S | 4.523 | 1.80 | 0.207 | 0.246 |
| Lag-Llama | 4.859 | 2.48 | 0.333 | 0.072 |
| Kronos | 4.498 | 2.10 | $10.23^{\dagger}$ | -5.953 |
| *Panel: $h = 22$ (22 days)* | | | | |
| HAR | 2.169 | 2.77 | $1.08^{\dagger}$ | 0.078 |
| HAR-J | 2.234 | 2.73 | $1.23^{\dagger}$ | 0.071 |
| HAR-RS | 2.201 | 2.89 | $4.01^{\dagger}$ | -0.034 |
| HARQ | 2.126 | 2.66 | $17.88^{\dagger}$ | -0.051 |
| Log-HAR | 2.001 | 2.16 | 0.280 | 0.130 |
| ARFIMA | 3.681 | 2.65 | 0.286 | -0.296 |
| Chronos-Bolt-S | 1.873 | 1.92 | 0.282 | 0.168 |
| Chronos-Bolt-B | 1.886 | 1.87 | 0.286 | 0.167 |
| Moirai-2.0-S | **1.855** | **1.83** | **0.238** | **0.201** |
| Lag-Llama | 2.062 | 2.29 | 0.336 | 0.082 |
| Kronos | 2.031 | 2.22 | $12.89^{\dagger}$ | -0.726 |

Table 6: Diebold–Mariano test: pairwise win rates. Each cell reports the percentage of pairwise comparisons (across 50 assets × 10 opponents) in which the row model achieves significantly lower QLIKE at the 5% level.

| Model | $h = 1$ | $h = 5$ | $h = 22$ |
|---|---|---|---|
| HAR | 40.8% | 23.0% | 7.4% |
| HAR-J | 24.0% | 20.2% | 7.0% |
| HAR-RS | 13.8% | 4.0% | 2.0% |
| HARQ | 0.6% | 1.0% | 0.4% |
| Log-HAR | 66.8% | 41.6% | 22.4% |
| ARFIMA | 30.0% | 17.8% | 19.4% |
| Chronos-Bolt-S | 51.8% | 36.4% | 17.8% |
| Chronos-Bolt-B | 51.2% | 32.8% | 15.8% |
| Moirai-2.0-S | 54.4% | 52.4% | 29.2% |
| Lag-Llama | 16.8% | 13.2% | 22.0% |
| Kronos | 13.0% | 8.6% | 1.6% |

Each cell reports the percentage of pairwise DM tests (across 50 assets × 10 opponents = 500 tests) in which the row model has significantly lower QLIKE at the 5% level.

## 5.1 Aggregate Forecast Accuracy

Table 2 reports the cross-sectional average of per-asset loss functions across 40 equities. Figure 1 illustrates the forecasts of three representative models against realized values for AAPL. The results reveal a systematic divergence between QLIKE and the other metrics.

At the one-day horizon ($h = 1$), Log-HAR achieves the lowest average QLIKE (0.220), followed closely by Moirai 2.0 (0.226) and Chronos-Bolt-Small (0.229). The gap between Log-HAR and the best TSFM is small—roughly 3% in relative terms—and is driven by the log-space formulation of Log-HAR, which guarantees positive forecasts and penalizes proportional forecast errors in a way that aligns with QLIKE's structure. On MSE, MAE, and $R^2_{\text{OOS}}$, however, the TSFMs dominate: Moirai 2.0 achieves the lowest MSE ($0.210 \times 10^{-6}$), and the highest $R^2_{\text{OOS}}$ (0.426), compared to 0.238 for Log-HAR. Chronos-Bolt-Base achieves the lowest MAE ($1.17 \times 10^{-4}$). The out-of-sample $R^2$ gap between TSFMs and the best econometric model is 8–19 percentage points at $h = 1$, a substantial margin.

At the five-day horizon ($h = 5$), the TSFM advantage becomes more pronounced.

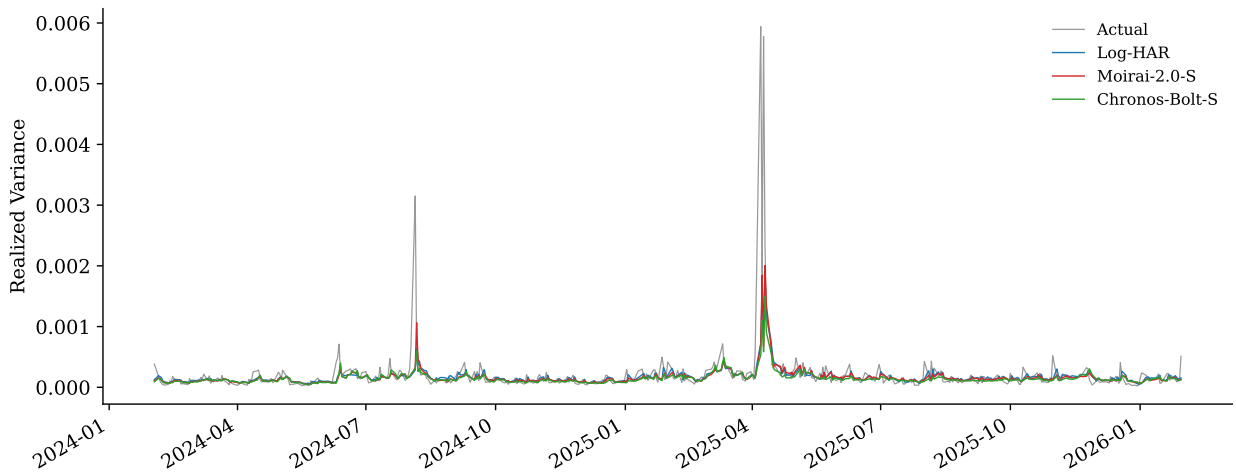


Figure 1: Forecast vs. actual realized variance for AAPL at $h = 1$. The last 500 out-of-sample observations are shown. Gray: actual RV; blue: Log-HAR; red: Moirai 2.0; green: Chronos-Bolt-Small.

Moirai 2.0 achieves the best score across all four metrics: MSE ($0.122 \times 10^{-6}$), MAE ($1.07 \times 10^{-4}$), QLIKE (0.188), and $R^2_{\text{OOS}}$ (0.384). The gap over Log-HAR in QLIKE terms is now 19%, compared to 3% at $h = 1$. Chronos-Bolt-Small achieves the second-lowest QLIKE (0.213), still below Log-HAR's 0.231. The econometric models that do best at $h = 1$ (HAR, HAR-J) deteriorate more rapidly at longer horizons, while the TSFMs maintain relatively stable accuracy.

At the monthly horizon ($h = 22$), Moirai 2.0 again dominates: QLIKE = 0.257, MSE = $0.098 \times 10^{-6}$, and $R^2_{\text{OOS}} = 0.110$. The econometric benchmarks now show near-zero or negative $R^2_{\text{OOS}}$, indicating that they barely outperform (or underperform) the historical mean at this horizon. Log-HAR achieves $R^2_{\text{OOS}} = -0.044$, while Chronos-Bolt-Small maintains a positive $R^2_{\text{OOS}}$ of 0.085. This horizon-dependent pattern is consistent with the hypothesis that TSFMs' large context windows (512 days) capture long-range dependencies more effectively than the HAR's fixed three-component structure.

Levels-based HAR extensions are unstable in this evaluation. HAR-RS and HARQ produce QLIKE values of 3.73 and 16.46 at $h = 1$, respectively, rising to 9.40 and 12.42 at $h = 22$. These inflated values arise because the OLS-estimated models occasionally pro-

duce near-zero forecasts for $RV_{t+h}$, which are floored at $10^{-6}$ for QLIKE evaluation. Across the full sample, no forecast is literally negative, but the floor binds for a small fraction of observations. Since QLIKE $= RV/\hat{RV} - \log(RV/\hat{RV}) - 1$, a forecast near zero generates an extremely large loss whenever the realized value is far from zero, and a handful of such observations can dominate the cross-sectional average. Table 12 (Appendix) reports cross-sectional medians, which are robust to these outliers; the qualitative ranking of models is unchanged. Log-HAR and ARFIMA avoid this problem entirely by operating in log space, where forecasts are positive by construction. TSFMs also produce strictly positive forecasts, since their training distributions are bounded below. This practical advantage may be as important as any differences in statistical accuracy: a model that occasionally produces degenerate forecasts is difficult to deploy in production risk systems.

Bootstrap confidence intervals (10,000 draws, percentile method) on the cross-sectional mean QLIKE confirm the statistical significance of the key rankings. At $h = 1$, the 95% CI for Log-HAR QLIKE is $[0.204, 0.239]$, overlapping with Moirai 2.0's $[0.209, 0.247]$ and Chronos-Bolt-Small's $[0.211, 0.251]$—consistent with the MCS finding that these models are statistically indistinguishable. At $h = 5$, Moirai 2.0's CI $[0.172, 0.209]$ lies entirely below Log-HAR's $[0.208, 0.257]$, confirming the TSFM advantage at this horizon. At $h = 22$, Moirai 2.0 $[0.238, 0.277]$ and Lag-Llama $[0.260, 0.306]$ both separate cleanly from Log-HAR $[0.294, 0.417]$.

## 5.2 Cross-Asset and Cross-Market Heterogeneity

Tables 4 and 5 extend the analysis to foreign exchange and futures markets, revealing cross-asset-class heterogeneity in model rankings.

**Foreign exchange.** The FX sample covers five major currency pairs with lower volatility levels ($\bar{RV} \approx 3 \times 10^{-5}$) and moderate persistence ($\bar{\rho}_1 = 0.52$). For FX, Chronos-Bolt consistently outperforms Moirai 2.0. At $h = 1$, Log-HAR achieves the lowest QLIKE (0.163),

with Chronos-Bolt-Small close behind (0.174), while Moirai 2.0 underperforms substantially (QLIKE = 0.329, $R^2_{\text{OOS}} = -0.008$). At $h = 5$, Log-HAR retains the QLIKE lead (0.120 vs. 0.128 for Chronos-Bolt-Small), but Chronos-Bolt achieves the best MSE and $R^2_{\text{OOS}}$ (0.353). At $h = 22$, Log-HAR again leads on QLIKE (0.147), while Chronos-Bolt achieves the best $R^2_{\text{OOS}}$ (0.250). Moirai 2.0's poor FX performance ($R^2_{\text{OOS}} < 0$ at all horizons) stands in contrast to its equity dominance and suggests that Moirai's training data or architecture is less suited to the smoother, lower-amplitude dynamics of currency volatility.

**Futures.** The futures sample displays the widest cross-asset heterogeneity in volatility dynamics. The E-mini S&P 500 (ES) has first-order autocorrelation of 0.779, while Gold (GC) shows near-zero autocorrelation ($\rho_1 = -0.001$). At $h = 1$, Log-HAR achieves the lowest QLIKE (0.193), followed by Chronos-Bolt-Small (0.213) and HAR (0.225). Moirai 2.0 ranks behind (0.289). At $h = 5$, Log-HAR retains the QLIKE lead (0.165), with Chronos-Bolt-Small at 0.185 and Chronos-Bolt-Base at 0.196. The TSFMs achieve the best MSE and $R^2_{\text{OOS}}$. At the monthly horizon, Moirai 2.0 recovers and achieves the lowest QLIKE (0.238), MSE, and MAE, while levels-based HAR variants produce inflated QLIKE (HAR: 1.08, HARQ: 17.88).

**Cross-asset patterns.** Three patterns emerge from the cross-asset comparison. First, Log-HAR is consistently the best or near-best econometric specification across all asset classes, confirming the practical value of the log transformation for ensuring forecast positivity and aligning with QLIKE's loss structure. Second, the relative ranking of TSFMs shifts across asset classes: Moirai 2.0 dominates equities (where persistence is high), while Chronos-Bolt performs better for FX and short-horizon futures (where dynamics are smoother or less persistent). Third, all levels-based HAR extensions produce inflated QLIKE for futures, where extreme volatility events in Natural Gas and Crude Oil expose the negative-forecast problem. These asset-class-dependent results highlight the importance of multi-market evaluations: a model that appears dominant on equities alone may not generalize to other asset

classes.

## 5.3 Statistical Significance and Model Confidence Sets

Table 3 reports Model Confidence Set inclusion rates for 40 equities, and Table 6 summarizes pairwise Diebold–Mariano test results across all 50 assets.

**MCS results for equities.** At $h = 1$, Log-HAR enters the MCS for 90% of assets, confirming its status as the single most reliable specification at the daily horizon. Moirai 2.0 enters for 77.5%, Chronos-Bolt-Small and HAR each for 42.5%. Among the additional TSFMs, Kronos enters for only 2.5% and Lag-Llama for 0%, indicating that these models are systematically outperformed at short horizons. Levels-based HAR variants are also largely excluded: HAR-RS enters for 2.5% and HARQ for 0%.

At $h = 5$, the picture shifts. Moirai 2.0 enters the MCS for 100% of assets, followed by Log-HAR (65%), HAR (42.5%), and Chronos-Bolt-Small (30%). Lag-Llama enters for only 5% and Kronos for 0%.

At $h = 22$, the ranking changes markedly. Moirai 2.0 enters for 97.5%, Lag-Llama for 95%, ARFIMA for 87.5%, Log-HAR for 75%, and Chronos-Bolt-Small for 37.5%. Lag-Llama's sharp improvement from 0% at $h = 1$ to 95% at $h = 22$ suggests that its lag-based tokenization captures long-range dependencies more effectively than short-term dynamics. Kronos remains weak (7.5%). The high ARFIMA inclusion rate at $h = 22$ is consistent with its long-memory parameterization: at the monthly horizon, fractional differencing captures slow mean reversion that the HAR's three fixed lags approximate less precisely.

**Diebold–Mariano pairwise tests.** Table 6 reports the fraction of pairwise DM tests in which each model achieves significantly lower QLIKE at the 5% level, aggregated across 50 assets and 10 opponents. At $h = 1$, Log-HAR wins the highest fraction of pairwise comparisons (66.8%), followed by Moirai 2.0 (54.4%), Chronos-Bolt-Small (51.8%), and HAR (40.8%). The TSFMs' strong showing on this directional metric—where each win requires

both statistical significance and lower loss—confirms that their QLIKE advantage over levels-based HAR variants is not driven by a few outlier assets. At $h = 5$ and $h = 22$, Moirai 2.0 achieves the highest win rate (52.4% and 29.2%, respectively), reflecting its broad dominance at longer horizons. Lag-Llama's win rate rises from 16.8% at $h = 1$ to 22.0% at $h = 22$, consistent with its improved MCS inclusion at longer horizons.

**Interpretation.** The MCS and DM test results jointly indicate that the best zero-shot TSFMs have achieved statistical parity with the best econometric benchmarks for realized volatility forecasting. At $h = 1$, Log-HAR remains the marginal leader on QLIKE, but Moirai 2.0 and Chronos-Bolt are statistically indistinguishable from it for most assets. At $h = 5$ and $h = 22$, Moirai 2.0 is the single model most consistently in the confidence set. However, the results also reveal substantial heterogeneity across TSFMs: Lag-Llama and Kronos are systematically excluded at short horizons, with Lag-Llama recovering only at $h = 22$ and Kronos performing poorly throughout. This heterogeneity suggests that architecture and pretraining data matter as much as the zero-shot paradigm itself. The practical implication is that TSFMs offer a viable alternative to HAR-family models, particularly for horizons beyond one day, but practitioners should evaluate multiple TSFM architectures rather than assuming uniform performance.

## 5.4 Portfolio Application: Realized Covariance Forecasting

The preceding analysis focused on univariate realized variance forecasting. We now extend the evaluation to a multivariate setting by examining whether covariance forecasts from TSFMs can improve portfolio allocation. We construct global minimum variance (GMV) portfolios using four approaches to covariance forecasting and compare the resulting realized portfolio variance.

**Covariance forecasting methods.** We consider four covariance forecasting approaches. (i) *Element-HAR*: a separate HAR model is fitted to each element of the realized covariance

matrix, treating variances and covariances as independent univariate series. (ii) *HAR-DRD*: the realized covariance matrix is decomposed as $\Sigma_t = D_t R_t D_t$, where $D_t$ is the diagonal matrix of realized volatilities and $R_t$ is the realized correlation matrix; HAR models are fitted separately to each element of $D_t$ and $R_t$, and the covariance forecast is reconstructed. This DRD decomposition ensures positive semi-definiteness by construction when the forecast correlation matrix is adjusted to be PSD. (iii–iv) *Chronos-Bolt-Small* and *Moirai 2.0*: zero-shot element-wise TSFM forecasts applied to each covariance pair, followed by PSD projection. We also include the $1/N$ equal-weight portfolio as a naive benchmark.

**Forex GMV results.** Table 7 reports the average realized variance of GMV portfolios constructed from each forecasting model, along with portfolio turnover and concentration metrics. For the 5-asset FX portfolio, all model-based approaches achieve substantial variance reduction relative to $1/N$. At $h = 1$, Chronos-Bolt-Small achieves the lowest average realized variance ($3.01 \times 10^{-6}$), a 44.9% reduction relative to $1/N$ ($5.46 \times 10^{-6}$). HAR-DRD is close behind ($3.11 \times 10^{-6}$, 43.0% reduction), while Element-HAR ($3.52 \times 10^{-6}$, 35.5%) and Moirai 2.0 ($3.55 \times 10^{-6}$, 35.0%) are somewhat less effective. At $h = 5$, Chronos-Bolt and HAR-DRD maintain their advantage (45.5% and 44.7% reduction, respectively), while Moirai 2.0 produces an unstable covariance forecast that results in higher realized variance than $1/N$. At $h = 22$, all methods except Moirai 2.0 produce comparable variance reductions (31–39%).

**Futures GMV results.** For the 5-contract futures portfolio (Table 7), Chronos-Bolt-Small achieves the lowest realized variance at $h = 1$ ($3.36 \times 10^{-5}$, a 68.8% reduction vs. $1/N$) and $h = 5$ ($3.55 \times 10^{-5}$, 67.2%). Moirai 2.0 performs worse, particularly at $h = 5$, where its realized variance exceeds $1/N$ by a factor of 3.5. This failure stems from Moirai's element-wise covariance forecasts being internally inconsistent: the assembled matrices are not positive semi-definite, and the eigenvalue floor used in PSD projection ($10^{-10}$) is too small relative to the matrix scale (typical eigenvalues $\sim 10^{-5}$). The resulting near-singular

matrices produce extreme portfolio weights on a handful of dates, which dominate the average realized variance. Chronos-Bolt does not suffer this problem because its forecasts produce matrices with minimum eigenvalues around $10^{-6}$, keeping condition numbers below 10.

**Stocks GMV results.** For the 40-stock equity portfolio (Table 7), the dimensionality problem becomes acute. Only HAR-DRD produces sensible portfolios, achieving a realized variance of $5.11 \times 10^{-5}$ at $h = 1$—a 29.4% reduction relative to $1/N$ ($7.24 \times 10^{-5}$). All element-wise approaches (Element-HAR, Chronos-Bolt, Moirai 2.0) fail catastrophically: their realized variances exceed $1/N$ by two to three orders of magnitude, with average daily turnover above 7 (Chronos-Bolt) and 50 (Moirai). The problem is that element-wise forecasting of 820 unique covariance pairs produces assembled matrices with extremely small eigenvalues. PSD projection with a floor of $10^{-10}$ yields near-singular matrices whose inverse concentrates portfolio weight on a few assets. The DRD decomposition avoids this by separately modeling volatilities and correlations, ensuring that the reconstructed matrix inherits the PSD structure of the correlation forecast. This result highlights a practical limitation: element-wise TSFM covariance forecasting does not scale to high-dimensional portfolios without structural constraints. To verify that this failure is not an artifact of the eigenvalue floor, we re-projected all stocks forecast matrices using a floor of $10^{-6}$ (six orders of magnitude larger than the default $10^{-10}$) and re-evaluated the GMV portfolios. The higher floor reduces turnover substantially—from 13.7 to 2.2 for Chronos-Bolt and from 52.7 to 7.2 for Moirai at $h = 1$—but realized variances remain 5–7 times above $1/N$ ($3.6 \times 10^{-4}$ and $5.0 \times 10^{-4}$ vs. $7.2 \times 10^{-5}$). HAR-DRD continues to outperform $1/N$ ($4.6 \times 10^{-5}$). The failure of element-wise approaches thus reflects a fundamental structural problem, not a tuning parameter.

**Turnover.** The $1/N$ portfolio has zero turnover by construction. Among model-based portfolios in the low-dimensional FX and futures settings, HAR-DRD achieves the lowest turnover (0.080 for FX at $h = 1$), followed by Moirai 2.0 (0.060) and Chronos-Bolt

(0.116). The DRD decomposition's smoothness advantage comes from its separate modeling of volatilities and correlations, which dampens spurious variation in off-diagonal elements.

Table 7: Global minimum variance portfolio performance. Avg RV is the average daily realized variance of the GMV portfolio ($\times 10^{-6}$). Turnover is the average daily absolute weight change. For stocks, element-wise approaches (Element-HAR, Chronos-Bolt, Moirai) produce near-singular matrices that cause extreme portfolio weights; their Avg RV values are orders of magnitude above $1/N$ and are omitted from the Std RV and Max Wt columns. Bold indicates the lowest Avg RV in each panel.

| Model | $h$ | Avg RV | Std RV | Turnover | Max Wt |
|---|---|---|---|---|---|
| *Forex (5 assets)* | | | | | |
| $1/N$ | 1 | 5.46 | 10.30 | — | — |
| Element-HAR | | 3.52 | 6.50 | 0.096 | 0.383 |
| HAR-DRD | | 3.11 | 4.50 | 0.080 | 0.370 |
| Chronos-Bolt-S | | **3.01** | 4.78 | 0.116 | 0.369 |
| Moirai-2.0-S | | 3.55 | 9.83 | 0.060 | 0.384 |
| $1/N$ | 5 | 5.73 | 6.48 | — | — |
| Element-HAR | | 3.58 | 5.45 | 0.096 | 0.383 |
| HAR-DRD | | 3.17 | 2.99 | 0.080 | 0.370 |
| Chronos-Bolt-S | | **3.12** | 3.69 | 0.090 | 0.369 |
| Moirai-2.0-S | | 21.72 | 1083.82 | 0.104 | 0.397 |
| $1/N$ | 22 | 5.43 | 4.32 | — | — |
| Element-HAR | | 3.71 | 5.89 | 0.096 | 0.383 |
| HAR-DRD | | 3.30 | 2.45 | 0.080 | 0.370 |
| Chronos-Bolt-S | | **3.29** | 3.00 | 0.074 | 0.370 |
| Moirai-2.0-S | | 3.52 | 6.06 | 0.046 | 0.382 |
| *Futures (5 contracts)* | | | | | |
| $1/N$ | 1 | 107.77 | 316.84 | — | — |
| Element-HAR | | 37.47 | 47.56 | 0.116 | 0.518 |
| HAR-DRD | | 35.31 | 35.42 | 0.129 | 0.523 |
| Chronos-Bolt-S | | **33.63** | 33.30 | 0.114 | 0.520 |
| Moirai-2.0-S | | 45.08 | 156.03 | 0.264 | 0.552 |
| $1/N$ | 5 | 107.88 | 207.81 | — | — |
| Element-HAR | | 39.89 | 83.62 | 0.116 | 0.518 |
| HAR-DRD | | 36.83 | 33.59 | 0.129 | 0.523 |
| Chronos-Bolt-S | | **35.46** | 32.45 | 0.088 | 0.514 |
| Moirai-2.0-S | | 376.26 | 19571.31 | 0.321 | 0.560 |
| $1/N$ | 22 | 108.35 | 151.33 | — | — |
| Element-HAR | | 42.66 | 50.69 | 0.116 | 0.518 |
| HAR-DRD | | 40.75 | 38.02 | 0.129 | 0.523 |
| Chronos-Bolt-S | | **39.88** | 44.59 | 0.065 | 0.503 |
| Moirai-2.0-S | | 59.63 | 670.30 | 0.154 | 0.515 |
| *Stocks (40 assets)* | | | | | |
| $1/N$ | 1 | 72.38 | 200.30 | — | — |
| HAR-DRD | | **51.10** | 124.55 | 0.788 | 0.159 |
| Chronos-Bolt-S | | $> 10^4$ | — | 13.695 | — |
| Element-HAR | | $> 10^4$ | — | 12.672 | — |
| Moirai-2.0-S | | $> 10^4$ | — | 52.690 | — |
| $1/N$ | 5 | 72.39 | 130.35 | — | — |
| HAR-DRD | | **52.81** | 81.06 | 0.789 | 0.159 |
| Chronos-Bolt-S | | $> 10^4$ | — | 7.639 | — |
| Element-HAR | | $> 10^4$ | — | 12.691 | — |
| Moirai-2.0-S | | $> 10^4$ | — | 48.779 | — |
| $1/N$ | 22 | 72.21 | 93.25 | — | — |
| HAR-DRD | | **57.94** | 80.69 | 0.790 | 0.159 |
| Chronos-Bolt-S | | $> 10^4$ | — | 7.710 | — |
| Element-HAR | | $> 10^4$ | — | 12.774 | — |
| Moirai-2.0-S | | $> 10^4$ | — | 51.283 | — |

Table 8: Covariance forecast accuracy by asset class. Avg Frobenius is the average normalized Frobenius norm of the forecast error matrix ($\|\hat{\Sigma}_t - \Sigma_t\|_F/N$). Diag QLIKE is the average QLIKE computed on diagonal (variance) elements. Bold indicates the best value in each panel×horizon cell.

| | | $h = 1$ | | $h = 5$ | | $h = 22$ | |
|---|---|---|---|---|---|---|---|
| | Model | Frob ($\times 10^{-5}$) | QLIKE | Frob ($\times 10^{-5}$) | QLIKE | Frob ($\times 10^{-5}$) | QLIKE |
| *Forex* | Element-HAR | 1.06 | 0.149 | 0.89 | 0.105 | 0.88 | 0.116 |
| | HAR-DRD | 0.96 | 0.133 | 0.79 | **0.092** | 0.79 | **0.112** |
| | Chronos-Bolt-S | **0.86** | **0.145** | **0.72** | 0.112 | **0.72** | 0.151 |
| | Moirai-2.0-S | 1.38 | 0.308 | 1.28 | 0.245 | 1.17 | 0.211 |
| *Futures* | Element-HAR | 21.64 | 0.227 | 20.04 | 0.183 | 20.46 | 0.214 |
| | HAR-DRD | 15.06 | 0.158 | **13.35** | **0.125** | **13.71** | **0.175** |
| | Chronos-Bolt-S | **14.87** | **0.173** | 13.74 | 0.160 | 14.87 | 0.260 |
| | Moirai-2.0-S | 15.20 | 0.228 | 13.88 | 0.181 | 14.41 | 0.217 |
| *Stocks* | Element-HAR | 7.22 | 0.196 | 6.18 | 0.171 | 6.46 | 0.212 |
| | HAR-DRD | 6.87 | 0.166 | 5.88 | **0.150** | **5.81** | **0.204** |
| | Chronos-Bolt-S | **6.72** | **0.176** | **5.84** | 0.181 | 5.89 | 0.303 |
| | Moirai-2.0-S | 7.48 | 0.175 | 6.37 | 0.160 | 6.20 | 0.237 |

**Covariance forecast accuracy.** Table 8 reports element-wise forecast accuracy across all three asset classes. Two metrics are computed: the normalized Frobenius norm of the forecast error matrix ($\|\hat{\Sigma}_t - \Sigma_t\|_F/N$) and the QLIKE loss on diagonal (variance) elements. The results show that TSFMs produce competitive covariance forecasts at the element level, even when the assembled matrices are ill-conditioned for portfolio optimization. For FX at $h = 1$, Chronos-Bolt achieves the lowest Frobenius norm ($0.86 \times 10^{-5}$) and near-best diagonal QLIKE (0.145 vs. 0.133 for HAR-DRD). For futures, Chronos-Bolt and HAR-DRD are comparable on Frobenius norm, with HAR-DRD leading on QLIKE at $h = 5$ and $h = 22$. For stocks, Chronos-Bolt achieves the lowest Frobenius norm at $h = 1$ and $h = 5$, while HAR-DRD leads on diagonal QLIKE. Moirai 2.0 is competitive on stocks (QLIKE 0.175 at $h = 1$, matching Chronos-Bolt) but weaker on FX (QLIKE 0.308). The disconnect between element-wise accuracy and portfolio performance—Chronos-Bolt's Frobenius norm is lower than HAR-DRD's for stocks, yet its portfolios fail—confirms that the problem lies in the spectral properties of the assembled matrix, not in the accuracy of individual element forecasts.

# 6 Robustness Checks

We assess the stability of our main findings along two dimensions: cross-dataset validation and the behavior of model rankings across the full set of results.

**Cross-dataset validation.** We replicate the univariate forecasting exercise using the CAPIRe dataset, which provides realized volatility for 29 DJIA stocks over 2003–2024 in annualized percentage units—a different scale, sample period, and estimation methodology than VOLARE. Tables 10 and 11 (Appendix) report the results.

The CAPIRe results confirm the main patterns observed with VOLARE. TSFMs achieve the lowest MSE and highest $R^2_{\text{OOS}}$ at all horizons: Chronos-Bolt-Small achieves $R^2_{\text{OOS}} = 0.465$ at $h = 1$ and 0.495 at $h = 5$, compared to 0.382 and 0.369 for Log-HAR. On QLIKE, Log-HAR is best at $h = 1$ (0.139 vs. 0.149 for Chronos-Bolt-Small), but at $h = 5$ and $h = 22$ Chronos-Bolt-Small achieves the lowest QLIKE (0.155 and 0.249, respectively). Model rankings between VOLARE and CAPIRe are broadly consistent, with Log-HAR and the TSFMs consistently at the top and levels-based HAR variants penalized by inflated QLIKE. The stability of these findings across two independent datasets with different time periods, asset universes, and RV estimation methodologies strengthens the conclusion that TSFMs provide competitive zero-shot performance.

**Sub-sample stability: pre- and post-COVID.** Table 9 splits the equity sample at March 1, 2020, and reports forecast accuracy separately for the pre-COVID (2015–2020) and post-COVID (2020–2026) periods. The sub-sample analysis reveals that TSFM performance is regime-robust, while levels-based econometric models are not.

In the pre-COVID period, Log-HAR achieves the lowest QLIKE at all horizons (0.206 at $h = 1$, 0.182 at $h = 5$, 0.321 at $h = 22$), though Chronos-Bolt-Base matches it on MSE and $R^2_{\text{OOS}}$ at $h = 1$. The pre-COVID period features lower average volatility and fewer extreme events, conditions that favor the parsimonious HAR specification.

In the post-COVID period, TSFMs dominate across all metrics and horizons. Moirai 2.0 achieves the lowest QLIKE at $h = 1$ (0.213 vs. 0.225 for Log-HAR), $h = 5$ (0.174 vs. 0.252), and $h = 22$ (0.168 vs. 0.335), while also achieving the highest $R^2_{\text{OOS}}$ (0.430, 0.378, 0.148). The levels-based HAR variants suffer extreme QLIKE inflation in the high-volatility regime: HARQ's QLIKE rises from 2.35 pre-COVID to 23.63 post-COVID at $h = 1$, because the near-zero forecast problem is exacerbated when realized values are large. Log-HAR is more stable across regimes than its levels-based counterparts but still falls behind the TSFMs on all metrics in the post-COVID sample. The TSFM heterogeneity pattern persists across sub-samples: Lag-Llama underperforms in both periods at $h = 1$ (QLIKE 0.433 pre-COVID, 0.385 post-COVID) but improves at longer horizons, while Kronos achieves the lowest MSE pre-COVID at $h = 1$ (0.082) but has inflated QLIKE in both periods due to systematic forecast bias. These results suggest that TSFMs' advantage is concentrated in volatile market conditions, where their flexible nonparametric forecasts adapt more effectively than the HAR's linear structure.

**Remaining directions.** VOLARE provides alternative realized measures including subsampled realized variance ($RV^{(5)}_{\text{SS}}$), realized kernels, and range-based estimators. Evaluating model rankings under alternative volatility proxies and varying TSFM context lengths (128 to 512 days) are directions for future work that would further establish the generality of our findings.

# 7 Conclusion

This paper evaluates zero-shot time series foundation models for realized volatility forecasting across 50 assets spanning U.S. equities, foreign exchange, and futures. We compare four TSFMs—Chronos-Bolt, Moirai 2.0, Lag-Llama, and Kronos—against six econometric benchmarks at three forecast horizons using the VOLARE dataset, with formal statistical testing via Diebold–Mariano tests and Model Confidence Sets.

Table 9: Sub-sample forecast accuracy: pre-COVID (2015–2020) and post-COVID (2020–2026) periods for 40 U.S. equities (VOLARE).

| Model | MSE ($\times 10^{-6}$) | MAE ($\times 10^{-4}$) | QLIKE | $R^2_{\text{OOS}}$ |
|---|---|---|---|---|
| *Panel: $h = 1$, Pre-COVID* | | | | |
| HAR | 0.089 | 0.984 | 0.494 | 0.223 |
| HAR-J | 0.090 | 0.976 | 0.395 | 0.204 |
| HAR-RS | 0.102 | 1.037 | 1.312$^{\dagger}$ | 0.158 |
| HARQ | 0.091 | 0.959 | 2.346$^{\dagger}$ | 0.175 |
| Log-HAR | 0.087 | 0.862 | **0.206** | 0.242 |
| ARFIMA | 0.123 | 1.148 | 0.328 | -0.008 |
| Chronos-Bolt-S | 0.086 | 0.834 | 0.235 | 0.247 |
| Chronos-Bolt-B | 0.084 | **0.823** | 0.234 | 0.275 |
| Moirai-2.0-S | 0.084 | 0.857 | 0.251 | 0.235 |
| Lag-Llama | 0.106 | 1.188 | 0.433 | 0.007 |
| Kronos | **0.082** | 0.895 | 0.954 | **0.287** |
| *Panel: $h = 1$, Post-COVID* | | | | |
| HAR | 0.328 | 1.684 | 1.809$^{\dagger}$ | 0.330 |
| HAR-J | 0.324 | 1.667 | 1.539$^{\dagger}$ | 0.345 |
| HAR-RS | 0.356 | 1.797 | 5.151$^{\dagger}$ | 0.259 |
| HARQ | 0.406 | 1.780 | 23.625$^{\dagger}$ | 0.091 |
| Log-HAR | 0.348 | 1.545 | 0.225 | 0.225 |
| ARFIMA | 0.392 | 1.896 | 0.312 | 0.147 |
| Chronos-Bolt-S | 0.285 | 1.363 | 0.226 | 0.410 |
| Chronos-Bolt-B | 0.281 | **1.353** | 0.228 | 0.422 |
| Moirai-2.0-S | **0.277** | 1.360 | **0.213** | **0.430** |
| Lag-Llama | 0.388 | 1.899 | 0.385 | 0.120 |
| Kronos | 0.309 | 1.527 | 1.166$^{\dagger}$ | 0.340 |
| *Panel: $h = 5$, Pre-COVID* | | | | |
| HAR | 0.050 | 0.972 | 0.421 | 0.029 |
| HAR-J | 0.052 | 0.966 | 0.427 | 0.018 |
| HAR-RS | 0.061 | 1.025 | 2.051$^{\dagger}$ | -0.098 |
| HARQ | 0.050 | 0.954 | 1.814$^{\dagger}$ | 0.036 |
| Log-HAR | **0.044** | 0.839 | **0.182** | 0.190 |
| ARFIMA | 0.063 | 1.014 | 0.263 | -0.063 |
| Chronos-Bolt-S | 0.047 | 0.791 | 0.241 | 0.168 |
| Chronos-Bolt-B | 0.046 | **0.782** | 0.240 | 0.187 |
| Moirai-2.0-S | 0.045 | 0.789 | 0.217 | 0.202 |
| Lag-Llama | 0.056 | 1.024 | 0.274 | 0.026 |
| Kronos | 0.045 | 0.842 | 0.700 | **0.229** |
| *Panel: $h = 5$, Post-COVID* | | | | |
| HAR | 0.221 | 1.643 | 3.877$^{\dagger}$ | 0.193 |
| HAR-J | 0.214 | 1.621 | 3.529$^{\dagger}$ | 0.219 |
| HAR-RS | 0.249 | 1.777 | 9.622$^{\dagger}$ | 0.077 |
| HARQ | 0.265 | 1.694 | 23.517$^{\dagger}$ | -0.008 |
| Log-HAR | 0.226 | 1.500 | 0.252 | 0.108 |
| ARFIMA | 0.227 | 1.672 | 0.237 | 0.167 |
| Chronos-Bolt-S | 0.170 | 1.256 | 0.200 | 0.348 |
| Chronos-Bolt-B | 0.172 | 1.261 | 0.204 | 0.337 |
| Moirai-2.0-S | **0.162** | **1.216** | **0.174** | **0.378** |
| Lag-Llama | 0.246 | 1.638 | 0.321 | 0.020 |
| Kronos | 0.195 | 1.421 | 0.953 | 0.220 |
| *Panel: $h = 22$, Pre-COVID* | | | | |
| HAR | 0.095 | 1.145 | 1.587$^{\dagger}$ | -0.032 |
| HAR-J | 0.095 | 1.140 | 1.510$^{\dagger}$ | -0.028 |
| HAR-RS | 0.101 | 1.197 | 3.796$^{\dagger}$ | -0.084 |
| HARQ | 0.095 | 1.134 | 2.323$^{\dagger}$ | -0.039 |
| Log-HAR | **0.091** | 1.062 | **0.321** | -0.006 |
| ARFIMA | 0.098 | 1.150 | 0.361 | -0.065 |
| Chronos-Bolt-S | 0.115 | 1.095 | 0.533 | -0.029 |
| Chronos-Bolt-B | 0.115 | 1.090 | 0.531 | -0.027 |
| Moirai-2.0-S | 0.113 | **1.057** | 0.427 | **-0.002** |
| Lag-Llama | 0.114 | 1.184 | 0.362 | -0.007 |
| Kronos | 0.117 | 1.178 | 1.016$^{\dagger}$ | -0.029 |
| *Panel: $h = 22$, Post-COVID* | | | | |
| HAR | 0.118 | 1.621 | 8.230$^{\dagger}$ | -0.070 |
| HAR-J | 0.117 | 1.608 | 8.084$^{\dagger}$ | -0.047 |
| HAR-RS | 0.133 | 1.739 | 11.923$^{\dagger}$ | -0.243 |
| HARQ | 0.139 | 1.637 | 17.488$^{\dagger}$ | -0.331 |
| Log-HAR | 0.116 | 1.470 | 0.335 | -0.102 |
| ARFIMA | 0.119 | 1.418 | 0.205 | -0.031 |
| Chronos-Bolt-S | 0.092 | 1.152 | 0.211 | 0.121 |
| Chronos-Bolt-B | 0.093 | 1.160 | 0.217 | 0.108 |
| Moirai-2.0-S | **0.089** | **1.084** | **0.168** | **0.148** |
| Lag-Llama | 0.105 | 1.285 | 0.239 | -0.012 |
| Kronos | 0.102 | 1.298 | 0.854 | 0.003 |

Three main findings emerge. First, the best zero-shot TSFMs are competitive with the best econometric benchmarks. This result differs from prior work on daily returns (Rahimikia et al., 2025) and on TimesFM 1.0 (Goel et al., 2025), and is consistent with the hypothesis that realized volatility's positive, persistent, mean-reverting dynamics align more closely with the macroeconomic and physical time series on which TSFMs were pretrained. However, not all TSFMs perform equally: Kronos is excluded from the MCS for virtually all assets at all horizons, and Lag-Llama is excluded at short horizons ($h = 1$, $h = 5$) but enters for 95% of assets at $h = 22$. Second, the ranking of models depends on the loss function and forecast horizon. Log-HAR achieves the lowest QLIKE at $h = 1$ for equities, but TSFMs dominate on MSE, MAE, and $R^2_{\text{OOS}}$ at all horizons. At $h = 5$ and $h = 22$, Moirai 2.0 achieves the lowest QLIKE as well. Model Confidence Set analysis shows that Moirai 2.0 enters the MCS for 100% of assets at $h = 5$ and 97.5% at $h = 22$, while Log-HAR enters for 90% at $h = 1$. Third, model rankings are not stable across asset classes: Chronos-Bolt outperforms on FX and futures, while Moirai 2.0 dominates equities.

A practical finding is that levels-based HAR extensions (HAR-RS, HARQ) produce near-zero forecasts that inflate QLIKE by orders of magnitude, a problem that log-space models and TSFMs avoid. Sub-sample analysis reveals that TSFM advantages are concentrated in the volatile post-COVID period (2020–2026), where Moirai 2.0 dominates all metrics and horizons, while Log-HAR retains a QLIKE edge in the calmer pre-COVID period. For portfolio construction, element-wise zero-shot TSFM forecasts of realized covariance matrices achieve substantial variance reductions (35–69%) relative to $1/N$ in low-dimensional GMV portfolios (5 assets), with Chronos-Bolt performing comparably to HAR-DRD baselines. However, element-wise TSFM covariance forecasting fails for the 40-stock portfolio, where only the structurally constrained HAR-DRD approach produces viable allocations—despite TSFMs achieving competitive element-wise forecast accuracy.

Several limitations should be noted. We evaluate only zero-shot TSFM performance; fine-tuning on realized volatility data may yield further gains. Our evaluation is restricted to

point forecasts; density forecast evaluation and interval calibration (Adler et al., 2025) remain open questions. We use a single volatility estimator (5-minute realized variance); sensitivity to the choice of estimator is left for future work. The covariance forecasting exercise uses a simple element-wise approach that fails at high dimensions; natively multivariate TSFM architectures (Ansari et al., 2025) or structural decompositions applied to TSFM outputs may address this limitation.

Future work should pursue four directions. First, parameter-efficient fine-tuning (e.g., LoRA) on realized volatility data may close the remaining gap with Log-HAR on QLIKE at short horizons. Second, density forecast evaluation would test whether TSFMs provide well-calibrated prediction intervals, a property required for Value-at-Risk and Expected Shortfall applications. Third, natively multivariate TSFMs could be applied directly to realized covariance matrices, potentially outperforming the element-wise approach used here. Fourth, extending the evaluation to emerging markets and alternative asset classes would test the generality of our findings.

The broader implication of our results is that the best zero-shot time series foundation models have reached a level of accuracy that makes them a practical tool for volatility forecasting. For practitioners, Moirai 2.0 and Chronos-Bolt combine competitive accuracy with zero estimation overhead and immediate deployment to new assets—advantages that become increasingly valuable as the number of assets to be forecast grows. At the same time, the poor performance of Lag-Llama and Kronos at short horizons, and the failure of element-wise TSFM covariance forecasting at high dimensions, caution against treating TSFMs as a universal solution. Model selection and structural constraints remain essential.

# A CAPIRe Dataset Results

Table 10: Forecast accuracy for 29 DJIA stocks (CAPIRe dataset). Values are cross-sectional averages. RV is in annualized % units. Bold indicates best value per column per panel. † denotes inflated QLIKE.

| Model | MSE | MAE | QLIKE | $R^2_{\text{OOS}}$ |
|---|---|---|---|---|
| *Panel: $h = 1$ (1 day)* | | | | |
| HAR | 24.059 | 1.08 | 0.474 | 0.383 |
| HAR-J | 24.213 | 1.08 | 0.512 | 0.373 |
| HAR-RS | 25.077 | 1.14 | 1.27† | 0.353 |
| HARQ | 29.702 | 1.11 | 12.57† | 0.247 |
| Log-HAR | 23.642 | 0.98 | **0.139** | 0.382 |
| ARFIMA | 27.555 | 1.29 | 0.276 | 0.281 |
| Chronos-Bolt-S | 21.150 | 0.91 | 0.149 | 0.465 |
| Chronos-Bolt-B | **20.940** | 0.90 | 0.148 | **0.466** |
| Moirai-2.0-S | 21.537 | **0.89** | 0.233 | 0.452 |
| *Panel: $h = 5$ (5 days)* | | | | |
| HAR | 14.718 | 1.14 | 1.28† | 0.371 |
| HAR-J | 14.883 | 1.13 | 1.28† | 0.364 |
| HAR-RS | 15.454 | 1.21 | 3.25† | 0.323 |
| HARQ | 18.231 | 1.16 | 13.26† | 0.230 |
| Log-HAR | 14.416 | 1.00 | 0.156 | 0.369 |
| ARFIMA | 15.626 | 1.19 | 0.252 | 0.306 |
| Chronos-Bolt-S | **11.932** | 0.89 | **0.155** | **0.495** |
| Chronos-Bolt-B | 12.142 | **0.88** | 0.156 | 0.483 |
| Moirai-2.0-S | 12.868 | 0.88 | 0.214 | 0.451 |
| *Panel: $h = 22$ (22 days)* | | | | |
| HAR | 13.853 | 1.40 | 5.29† | 0.085 |
| HAR-J | 14.051 | 1.40 | 5.25† | 0.070 |
| HAR-RS | 15.030 | 1.48 | 7.82† | -0.025 |
| HARQ | 318.153 | 1.49 | 12.88† | -12.386 |
| Log-HAR | 12.829 | 1.23 | 0.260 | 0.164 |
| ARFIMA | 12.789 | 1.20 | 0.286 | 0.090 |
| Chronos-Bolt-S | **10.846** | 1.01 | **0.249** | **0.308** |
| Chronos-Bolt-B | 11.069 | 1.00 | 0.255 | 0.289 |
| Moirai-2.0-S | 11.327 | **0.99** | 0.292 | 0.270 |

Table 11: Model Confidence Set inclusion rates for 29 DJIA stocks (CAPIRe dataset). Format as Table 3.

| Model | $h = 1$ (%) | $h = 5$ (%) | $h = 22$ (%) |
|---|---|---|---|
| HAR | 3.4 | 44.8 | 86.2 |
| HAR-J | 6.9 | 48.3 | 86.2 |
| HAR-RS | 3.4 | 27.6 | 72.4 |
| HARQ | 0.0 | 27.6 | 75.9 |
| Log-HAR | 96.6 | 96.6 | 100.0 |
| ARFIMA | 0.0 | 27.6 | 93.1 |
| Chronos-Bolt-S | 44.8 | 86.2 | 93.1 |
| Chronos-Bolt-B | 58.6 | 86.2 | 93.1 |
| Moirai-2.0-S | 51.7 | 75.9 | 93.1 |

# B Median Aggregation

Table 12 reports cross-sectional medians rather than means for the 40-equity sample, providing robustness to outlier assets with degenerate forecasts (e.g., HARQ at $h = 22$).

Table 12: Forecast accuracy for 40 U.S. equities (VOLARE), cross-sectional medians. Median aggregation is robust to outlier assets with degenerate forecasts. Bold indicates the best value in each column within each panel.

| Model | MSE ($\times 10^{-6}$) | MAE ($\times 10^{-4}$) | QLIKE | $R^2_{\text{OOS}}$ |
|---|---|---|---|---|
| *Panel: $h = 1$ (1 day)* | | | | |
| HAR | 0.116 | 1.14 | 0.221 | 0.352 |
| HAR-J | 0.113 | 1.14 | 0.567 | 0.365 |
| HAR-RS | 0.124 | 1.20 | $1.37^{\dagger}$ | 0.318 |
| HARQ | 0.173 | 1.24 | $11.12^{\dagger}$ | 0.066 |
| Log-HAR | 0.139 | 1.12 | **0.208** | 0.237 |
| ARFIMA | 0.147 | 1.37 | 0.297 | 0.163 |
| Chronos-Bolt-S | 0.096 | 0.97 | 0.212 | 0.436 |
| Chronos-Bolt-B | 0.094 | **0.95** | 0.213 | **0.459** |
| Moirai-2.0-S | **0.091** | 0.98 | 0.208 | 0.451 |
| Lag-Llama | 0.156 | 1.38 | 0.384 | 0.134 |
| Kronos | 0.112 | 1.08 | 0.413 | 0.379 |
| *Panel: $h = 5$ (5 days)* | | | | |
| HAR | 0.096 | 1.13 | 0.269 | 0.253 |
| HAR-J | 0.094 | 1.12 | 0.384 | 0.271 |
| HAR-RS | 0.109 | 1.19 | $3.59^{\dagger}$ | 0.179 |
| HARQ | 0.126 | 1.17 | $13.39^{\dagger}$ | 0.017 |
| Log-HAR | 0.111 | 1.08 | 0.205 | 0.133 |
| ARFIMA | 0.098 | 1.19 | 0.232 | 0.192 |
| Chronos-Bolt-S | 0.077 | 0.90 | 0.196 | 0.366 |
| Chronos-Bolt-B | 0.080 | 0.91 | 0.196 | 0.359 |
| Moirai-2.0-S | **0.073** | **0.86** | **0.172** | **0.383** |
| Lag-Llama | 0.128 | 1.20 | 0.295 | 0.039 |
| Kronos | 0.099 | 1.02 | 0.295 | 0.246 |
| *Panel: $h = 22$ (22 days)* | | | | |
| HAR | 0.078 | 1.23 | $3.40^{\dagger}$ | 0.007 |
| HAR-J | 0.076 | 1.22 | $3.39^{\dagger}$ | 0.016 |
| HAR-RS | 0.082 | 1.30 | $5.23^{\dagger}$ | -0.090 |
| HARQ | 0.089 | 1.25 | $9.29^{\dagger}$ | -0.121 |
| Log-HAR | 0.081 | 1.13 | 0.296 | -0.043 |
| ARFIMA | 0.075 | 1.10 | 0.269 | 0.073 |
| Chronos-Bolt-S | 0.071 | 0.94 | 0.319 | 0.085 |
| Chronos-Bolt-B | 0.072 | 0.97 | 0.326 | 0.082 |
| Moirai-2.0-S | **0.069** | **0.89** | **0.248** | **0.112** |
| Lag-Llama | 0.079 | 1.03 | 0.267 | 0.002 |
| Kronos | 0.078 | 1.06 | 0.542 | 0.024 |

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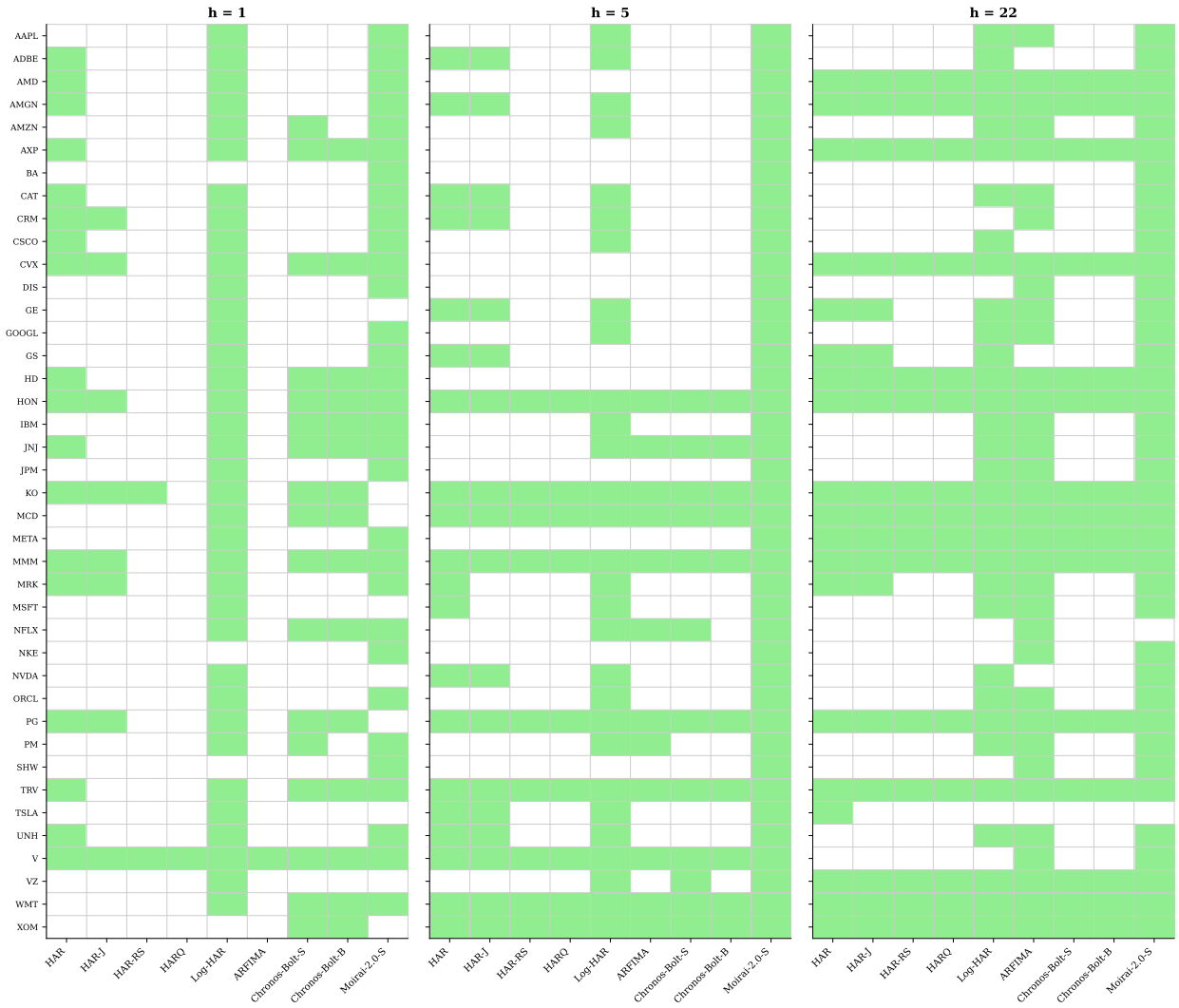


Figure 2: Model Confidence Set inclusion for 40 U.S. equities. Green cells indicate that the model is included in the MCS at the 10% significance level (QLIKE loss). Left: $h = 1$; center: $h = 5$; right: $h = 22$.

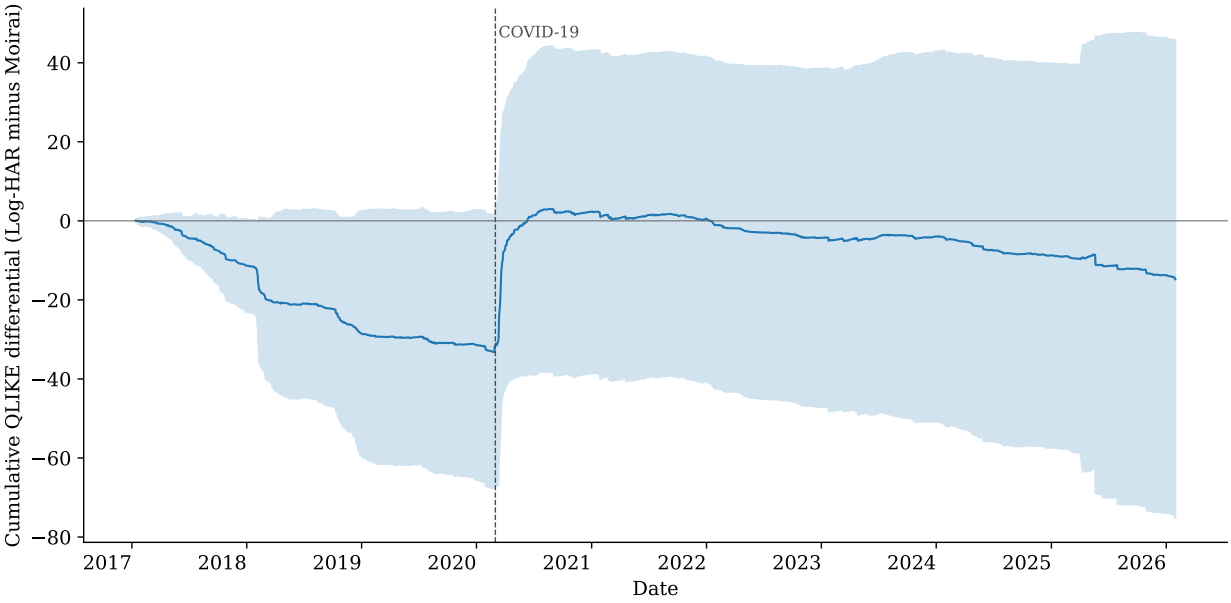


Figure 3: Cumulative QLIKE loss differential (Log-HAR minus Moirai 2.0) at $h = 1$, averaged across 40 equities. Positive values indicate that Moirai 2.0 outperforms Log-HAR. The shaded area is $\pm 1$ cross-sectional standard deviation. The vertical dashed line marks March 2020 (COVID-19 onset).